				[2]		
12	**(a)**		$H_0 : \mu = 1.5$ $H_1 : \mu < 1.5$	**B1**	**1.1**	both hypotheses in terms of μ
			μ is the **population mean** weight of flour in a bag	**B1**	**2.5**	allow μ is the **population mean** weight of a bag of flour
				[2]		

Question			Answer	Marks	AO	Guidance
12	(b)		$N(a, b)$ $a = 1.5$ or $b = \frac{0.24^2}{32}$ or 0.0018 $N(1.5, \frac{0.24^2}{32})$ **isw** or $N(1.5, 0.0018)$ **isw**	**M1** **A1** **A1**	**3.3** **2.2a** **3.1a**	a and b are numerical values allow 0.0424^2 for variance **A0** if answer spoiled by wrong variable quoted eg $\mu \sim N(1.5, \frac{0.24^2}{32})$ or $X \sim N(1.5, \frac{0.24^2}{32})$; allow only $\bar{X}$ **oe** if variable included
				[3]		
12	(c)		$0.0786 > 0.05$ or $-1.4142 > -1.645$ do not reject H_0 there is **insufficient evidence** at the 5% level to **suggest** that the **mean** weight of the flour in the bags is **less** than 1.5 kg	**M1** **A1** **A1**	**3.4** **1.1** **2.2b**	**or** $1.44 > 1.43(02586 \ldots)$ **NB** 1.43(02586…) is from InvNorm(0.05, 1.5,0.0424) allow accept H_0 **or** not significant **or** reject H_1 do not allow eg conclude / prove / indicate or other assertive statement instead of suggest if calculated values are used full marks may be awarded for awrt 0.07865 or 0.0786, **or** $-1.415 \leq z \leq -1.414$; otherwise award a maximum of **M1A1** for 0.07…or –1.4… other calculated values score **M0**
				[3]		